PathwayGAT: an explainable GNN framework integrating multi-layer information to reveal novel insights of complex biological processes

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Abstract

Graph Neural Networks (GNNs) have emerged as a powerful tool for biological research, particularly in modeling interactions between genes in various biological processes. However, there is still a significant gap in studies integrating multi-layered biological information beyond genes, such as microbes and SNPs into a GNN framework. This integration is crucial, as research has shown that these components could play critical roles in biological processes. Ignoring them can lead to an incomplete understanding of the underlying mechanisms driving complex biological phenomena. To address this gap, we developed an approach that constructs a comprehensive network encompassing genes, microbes, SNPs, potentially including other components, organized into interconnected pathways. After training our GNN-based model on this network with multiple datasets, we obtained satisfying results on different tasks. Besides, our framework, PathwayGAT, is also explainable. Beyond its success in various predictive and analytical tasks, PathwayGAT's explainable framework enables a detailed examination of each feature within the network, shedding light on their individual and collective contributions to the biological process under study. This feature not only enhances the model's utility but also makes it a valuable resource for researchers aiming to investigate complex biological systems further. PathwayGAT provides a robust platform for revealing mechanisms underlying biological processes, paving the way for novel insights and discoveries in computational biology.

Introduction

Graph Neural Networks (GNNs) have been widely used to explore the regulatory dynamics of various biological processes[1,2]. However, most studies have focused solely on genes within their constructed networks, overlooking valuable information from other biological components such as microbes and SNPs. Previous research has demonstrated that these components can play critical roles in diverse biological processes[3,4]. Neglecting them limits our ability to uncover the true mechanisms underlying these systems. In this study, we constructed pathway-based networks integrating information from genes, microbes, and SNPs. We then trained GNN models on these networks to perform a range of tasks. The model demonstrated robust performance, but more importantly, the explainability of the model allowed us to assess the individual contributions of each node and feature within the network. This transparency facilitated a deeper understanding of the biological significance of each element and how they collaborate to execute complex biological functions. We believe that PathwayGAT, our proposed framework, holds great potential for researchers investigating intricate biological systems, as it is a powerful tool to uncover the mechanisms driving these processes and advancing our knowledge toward them.

Methods

We first constructed the network with pathways. All pathways and annotations came from WikiPathways[5]. Each pathway served as a node in the network, while genes, microbes, SNPs and other components would be features of each node. The value of each feature would be set to zero if the corresponding component is not proven to be associated with the pathway. By integrating information across the network through GAT layers, the model could perform multiple kinds of tasks with neural network on top of the network embedding obtained from GNN. After validating the performance of our model, GNNExplainer[6] is used to explain the model, while the node mask is extracted for further study.

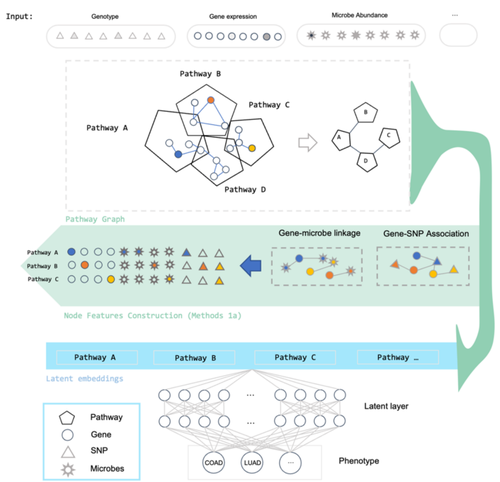

Fig.1 Schematic of PathwayGAT model

Results

We first tested our model's performance on TCGA dataset, where the model would distinguish between cancer samples and normal samples with gene and microbe data. The model performed well on both cancer types, while slightly better on COAD. Meanwhile, as we studied important elements identified by the model, we were able to retrieve a range of cancer-related pathways, genes, and microbes as proven by previous studies. Moreover, the interaction between them was captured by the model, including between pathways and features, as well as interaction between genes and microbes. Overall, the model reached adequate performance for the task and identified important elements in two types of cancer.

Results

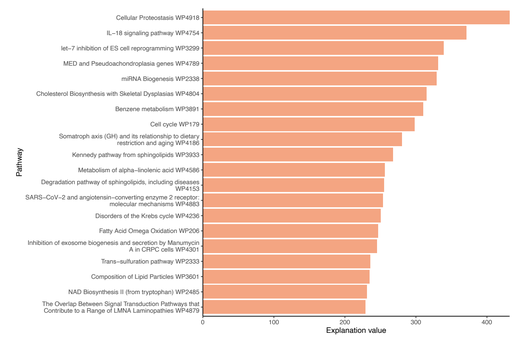

Fig.2 Important pathways identified by model on TCGA-COAD
Among all the pathways identified by our model, a lot of them were related to cancer.

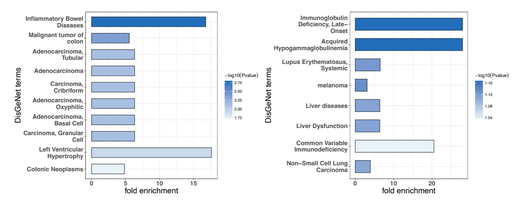

Fig.3 DisGeNet enrichment of top-ranking genes of COAD (left) and LUAD (right) model
The top-ranking genes were enriched in COAD and LUAD pathways, as annotated by DisGeNet[7].

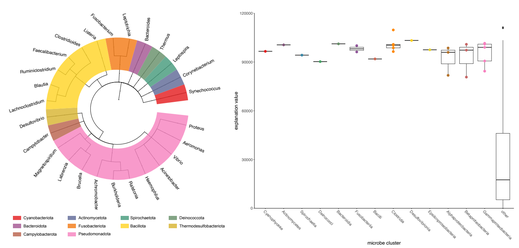

Fig.4 Phylogeny of important microbes identified by our model on COAD, and each phylum's importance compared with other microbes

Results

We then evaluated our model on distinguishing COAD and LUAD samples with genes and SNPs. We found that adding SNPs generally improved the performance of the model. We also found that top-ranking SNP associated genes were more related to cancer than top-ranking genes in the network, emphasizing the importance of adding SNP to the analysis.

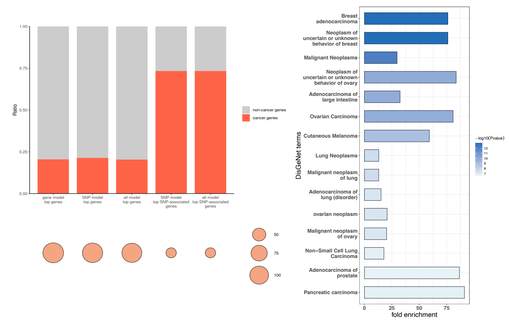

Fig.5 Result on SNP models
Top left is a bar plot where red bar shows the percentage of cancer-related genes according to NCG[8], identified in either top genes (first three bars) or top SNP associated genes (last two bars). Bottom left is a bubble plot showing the relative size of different gene sets. On the right side is a bar plot showing enriched pathway of DisGeNet[7] with top SNP-associated genes. Many of the enriched pathways are cancer-related pathways.

References

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